

VARDHAMAN COLLEGE OF ENGINEERING

DEPARTMENT OF CSE

Mini Project Review - II

Title: A Trust-Layered Framework for Discovering Authentic Student Opportunities.

Team No: 23MPCS-H8

S.No	Roll No	Student Name
1	23881A05EX	B. Navtej Reddy
2	23881A05FY	K. Ravi Varma
3	23881A05FM	G. Nandini

Guide Details

Ms. Ruqqaiya Begum
Assistant Professor

Table of Contents

1. Title
2. Background Research
3. Project Planning (Include Gantt Chart)
4. Literature Survey
5. Objectives
6. Project Milestone Tracker
7. Q&A

TITLE

- **Project Title:** A Trust-Layered Framework for Discovering Authentic Student Opportunities.
- **Domain/Area:** Web Development ,Artificial Intelligence, Machine Learning
- **Type:** Application + Research
- **Keywords:** Opportunity Verification,Trust-Layered Framework,AI-Based Recommendation System,Student Opportunity Platform,Fraud Detection,Skill–Opportunity Matching

BACKGROUND RESEARCH

Existing Solutions

- **Job & Internship Portals (LinkedIn, Internshala, Indeed):** Provide listings for internships and jobs, but do not verify opportunities at the institutional level and often contain outdated or misleading posts.
- **Scholarship & Hackathon Platforms (Unstop, Devpost, HackerEarth):** Aggregate events and scholarships, but lack cross-college validation and personalized matching based on student skills.
- **College Management Systems (CMS / ERP Portals):** Share internal opportunities within a single institution, but do not support inter-college collaboration or external opportunity validation.

BACKGROUND RESEARCH

Challenges

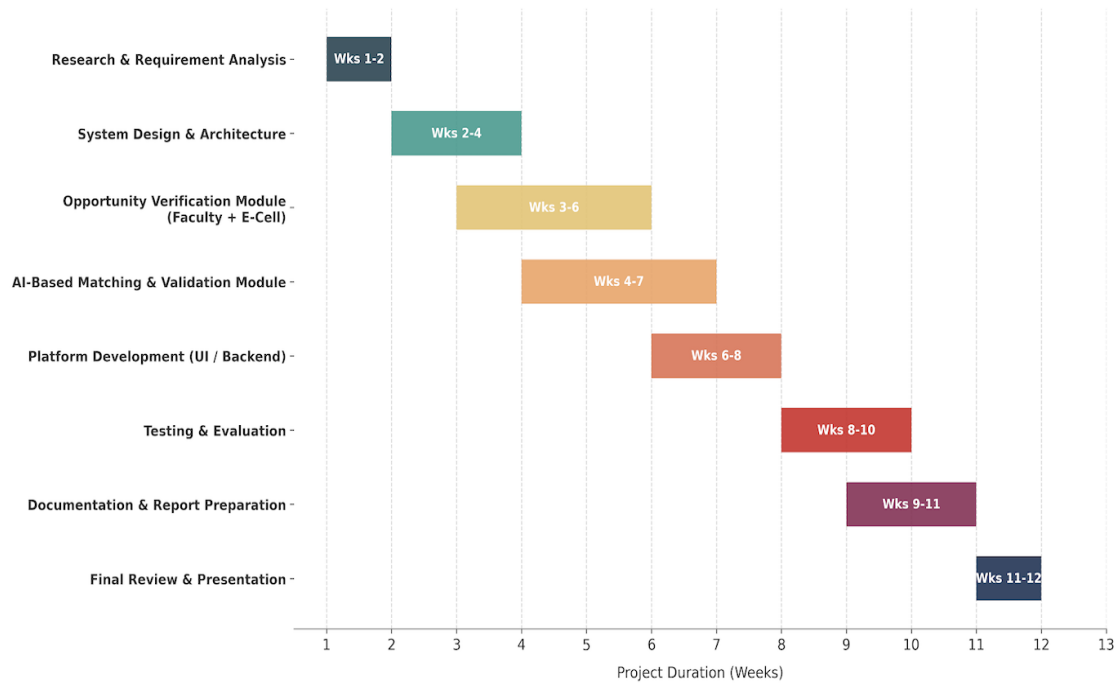
- **Lack of Authenticity Verification:** Most platforms do not verify the legitimacy of opportunities, leading to fake, misleading, or scam postings.
- **Irrelevant Opportunity Recommendations:** Existing systems provide generic listings without accurately matching student skills, interests, or career goals.
- **Absence of Institutional Trust Layer:** There is no faculty or institution-backed validation, limiting recruiter confidence and inter-college collaboration.
- **Fragmented Information Sources:** Opportunities are scattered across multiple platforms, making it difficult for students to track and apply on time.

BACKGROUND RESEARCH

Motivation

- **Need for a Trusted Opportunity Platform:** To provide students with access to only verified and authentic hackathons, internships, and scholarships.
- **Reducing Misinformation and Scams:** To eliminate fake, duplicate, and expired opportunities through institutional and AI-based verification.
- **Strengthening Institutional Credibility:** To build a faculty- and E-Cell-backed system that increases trust among students, colleges, and recruiters.
- **Encouraging Innovation and Participation:** To improve student engagement in real-world innovation, research, and career-building activities.

PROJECT PLANNING



LITERATURE SURVEY

Paper Title	Approach/Algorithm	Dataset Used	Key Findings	Gaps Identified
Fake Job Posting Detection Using Machine Learning	NLP + Random Forest, Logistic Regression	https://www.kaggle.com/shivamb/real-or-fake-fake-jobposting-prediction	ML models effectively detect fraudulent job listings	Focused only on job posts, not student opportunities
Recommender Systems in Practice	Collaborative Filtering	https://grouplens.org/datasets/movielens/1m/	Personalized recommendations are effective	Cold-start problem
AI-Based Skill Matching in Recruitment	NLP + Similarity Scoring	https://github.com/jneidel/job-descriptions	Improves candidate–job relevance	Bias and fairness issues
Digital Credential Verification Using Blockchain	Distributed Ledger Technology	https://www.opencerts.io/	Prevents academic certificate forgery	Infrastructure overhead

OBJECTIVES

To design and develop a centralized platform that provides B.Tech students with access to authentic hackathons, internships, and scholarship opportunities.

To implement a dual-layer verification mechanism involving faculty members and Entrepreneurship Cells (E-Cells) to ensure the authenticity and credibility of all posted opportunities.

To apply AI-based techniques for detecting fake, duplicate, and expired opportunity listings using text analysis and validation methods.

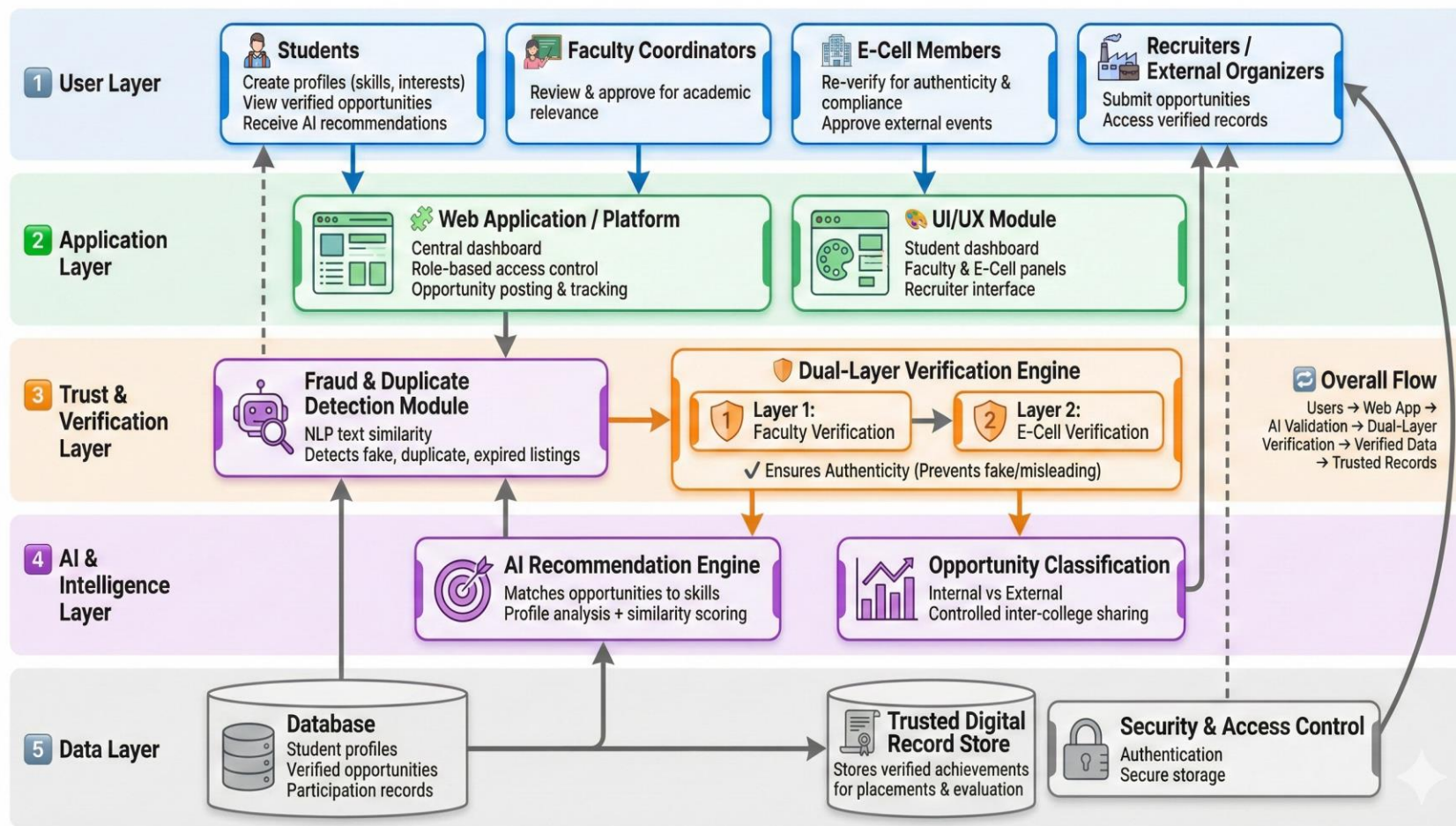
To develop an AI-driven recommendation system that matches students with relevant opportunities based on their skills, interests, academic background, and career goals.

To enable secure inter-college collaboration by categorizing opportunities as internal and external while maintaining institutional trust boundaries.

To create a trusted digital record of student participation and achievements that can be used confidently for placements, academic evaluation, and future references.

System Architecture

Project Architecture: Trust-Layered Framework for Authentic Student Opportunities



Tech stack Required

Category	Technology Stack
Frontend	 React.js & Tailwind CSS  
Backend	 Python (Flask)  
AI/ML Engine	 scikit-learn & TensorFlow  
Database	 PostgreSQL 
DevOps & Deployment	 Docker & AWS  

PROJECT MILESTONE TRACKER

- Paste Scanned Copy of Handbook Page No:3(Signed)

Q&A